PRACTICAL OBSERVATION CHECKLIST

ABERDARE TECHNICAL COLLEGE

ICT LEVEL 5 | MODULE 1 | PRACTICAL 1

UNIT: COMPUTER ESSENTIALS | UNIT CODE: 0611 351 01A

Time: 2 Hours | Practical: 50 Marks | Product: 5 Marks | Oral: 30 Marks | Grand Total: 85 Marks

Section A: Candidate and Assessor Details

Candidate Details	
Candidate Name	
Registration / Admission No.	
Course / Level	ICT Level 5
Class / Group	
Unit Name and Code	Computer Essentials - 0611 351 01A
Workstation No.	
Date of Assessment	
Start Time / End Time	

Assessor Details	
Assessor Name	
Staff / Registration No.	
Department	INFORMATION COMMUNICATION TECHNOLOGY

Assessor Details	
Assessor Signature	
Date	

Section B: Cutting List (Tools, Materials and Equipment Required)

The assessor confirms that the items below are available at each workstation before the assessment begins. This is a resource list for the practical; no items are cut.

S/No.	Item	Specification	Qty per Candidate
1	Desktop computer (system unit)	Working, case openable, with RAM, HDD/SSD, fan, expansion card/optical drive installed	1
2	Monitor	With power and display cable (VGA/HDMI)	1
3	Keyboard	USB or PS/2	1
4	Mouse	USB or PS/2	1
5	Screwdriver set	Phillips and flat-head, assorted sizes	1 set
6	Anti-static wrist strap	With earthing clip	1
7	Soft cloth	Lint-free	1
8	Soft brush	Anti-static	1
9	Compressed air can	Dust-blower	1
10	Printer	With power cable, USB/interface cable and driver software	1
11	Speakers	With 3.5 mm audio cable and power	1 pair
12	Projector (or other third peripheral)	With VGA/HDMI cable and power cable	1

S/No.	Item	Specification	Qty per Candidate
13	Power cables and extension socket	Earthed	As required
14	Printer paper	A4	5 sheets
15	Small container / anti-static tray	For holding screws and removed parts	1
16	Labeled peripheral devices and cables	For Task 1 identification	1 set
17	Stopwatch / wall clock	For timing	1
18	Assessment paper and this checklist	Printed	1 each

Section C: Practical Observation Checklist (50 Marks)

Instruction to assessor: award the full mark only where the step is performed correctly without help. Award 0 where the step is not done or is done incorrectly. Record observations under remarks.

Task 1: Identification of Computer Components and Ports (10 Marks)

No.	Observation Item	Max Marks	Marks Awarded	Remarks
1.1	Correctly points to and names component 1 inside the case (e.g. motherboard)	1		
1.2	Correctly points to and names component 2 (e.g. CPU / processor)	1		
1.3	Correctly points to and names component 3 (e.g. RAM)	1		
1.4	Correctly points to and names component 4 (e.g. power supply unit)	1		
1.5	Correctly points to and names component 5 (e.g. hard disk / SSD, cooling fan)	1		
1.6	Identifies rear port 1 and states a device that connects to it (e.g. VGA/HDMI - monitor)	1		
1.7	Identifies rear port 2 and states a device that connects to it (e.g. USB - keyboard/mouse/printer)	1		
1.8	Identifies rear port 3 and states a device that connects to it (e.g. audio jack - speakers)	1		
1.9	Identifies rear port 4 and states a device that connects to it (e.g. Ethernet/RJ-45 - network cable)	1		
1.10	Identifies rear port 5 and states a device that connects to it (e.g. PS/2 or power socket - keyboard/mouse or power cable)	1		

No.	Observation Item	Max Marks	Marks Awarded	Remarks
	Task 1 Sub-total	**10**		

Task 2: Disassembling and Cleaning Computer Devices (10 Marks)

No.	Observation Item	Max Marks	Marks Awarded	Remarks
2.1	Shuts down and disconnects power and all cables before opening the case	1		
2.2	Wears and correctly attaches the anti-static wrist strap before touching internal parts	1		
2.3	Selects and uses the correct screwdriver/tools	1		
2.4	Safely opens the case and keeps screws in a container	1		
2.5	Removes component 1 (named by examiner) using safe handling	1		
2.6	Removes component 2 (named by examiner) using safe handling	1		
2.7	Removes component 3 (named by examiner) using safe handling	1		
2.8	Cleans the case interior appropriately (brush / compressed air)	1		
2.9	Cleans the fan appropriately (holds blades while blowing air)	1		
2.10	Cleans removed components appropriately (e.g. soft cloth, no liquids on parts)	1		
	Task 2 Sub-total	**10**		

Task 3: Assembling Computer Hardware Devices (10 Marks)

No.	Observation Item	Max Marks	Marks Awarded	Remarks
3.1	Reinstalls component 1 in its correct slot/bay	1		
3.2	Reinstalls component 2 in its correct slot/bay	1		
3.3	Reinstalls component 3 in its correct slot/bay	1		
3.4	Secures components and reconnects internal power/data cables correctly	1		
3.5	Connects the monitor to the correct port	1		
3.6	Connects the keyboard to the correct port	1		
3.7	Connects the mouse to the correct port	1		

No.	Observation Item	Max Marks	Marks Awarded	Remarks
3.8	Closes and secures the case properly (no loose or leftover screws)	1		
3.9	Manages cables neatly (tied / routed, not strained)	1		
3.10	Follows safety precautions throughout assembly (no force, anti-static kept on until case closed)	1		
	Task 3 Sub-total	**10**		

Task 4: Booting the Computer (10 Marks)

No.	Observation Item	Max Marks	Marks Awarded	Remarks
4.1	Checks connections and powers on from a completely off state (cold boot)	1		
4.2	Performs the cold boot correctly until the operating system loads	1		
4.3	Observes and reports the screen/beep indicators during start-up	1		
4.4	Explains that POST (Power-On Self-Test) checks hardware such as CPU, RAM, keyboard and storage	1		
4.5	Explains the result of POST (passes then loads OS; errors shown as beep codes or messages)	1		
4.6	Explains the role of BIOS/UEFI in locating the boot device and loading the OS	1		
4.7	Performs a warm boot using an appropriate method (Ctrl+Alt+Del, Restart option or reset button)	1		
4.8	Warm boot completes with the system restarting correctly	1		
4.9	States one difference between cold and warm booting (e.g. cold boot starts from power off and runs full POST; warm boot restarts without cutting power)	1		
4.10	Uses correct terminology and logical sequence throughout	1		
	Task 4 Sub-total	**10**		

Task 5: Connecting and Configuring Peripheral Devices (10 Marks)

No.	Observation Item	Max Marks	Marks Awarded	Remarks
5.1	Connects the printer correctly (power and interface cable)	1		
5.2	Installs/selects the printer driver and sets it as default	1		

No.	Observation Item	Max Marks	Marks Awarded	Remarks
5.3	Prints a test page successfully (basic configuration)	1		
5.4	Connects the speakers to the correct audio port and power	1		
5.5	Adjusts volume and confirms audio output	1		
5.6	Connects the third peripheral (e.g. projector) with the correct cable	1		
5.7	Selects the correct input source / display mode (duplicate or extend)	1		
5.8	Adjusts resolution / display settings so the image is clear	1		
5.9	Confirms that the third peripheral works as required	1		
5.10	Observes safety (powers off before connecting, handles cables carefully)	1		
	Task 5 Sub-total	**10**		

Practical Checklist Summary

Task	Max Marks	Marks Awarded
Task 1: Identification of Components and Ports	10	
Task 2: Disassembling and Cleaning	10	
Task 3: Assembling Hardware	10	
Task 4: Booting the Computer	10	
Task 5: Connecting and Configuring Peripherals	10	
Total Practical Marks	**50**	

Section D: Product Checklist (5 Marks)

The assessor evaluates the final assembled and configured system after the candidate completes all tasks.

No.	Product Quality Item	Max Marks	Marks Awarded	Remarks
P1	System unit is fully assembled, case closed and secured with no loose parts or screws	1		
P2	Computer boots successfully to the desktop/login screen	1		
P3	Monitor, keyboard and mouse are properly connected and working	1		
P4	Printer, speakers and third peripheral are connected and working (test page, audio, display)	1		

No.	Product Quality Item	Max Marks	Marks Awarded	Remarks
P5	Cables are neat and the workstation is clean and tidy, with tools returned	1		
	Total Product Marks	**5**		

Section E: Oral Checklist (30 Marks)

Instruction to assessor: ask each question orally and award marks according to the guide. Award partial marks (1 to 2) for partially correct answers. Ten questions at 3 marks each.

No.	Oral Question	Expected Answer (Marking Guide)	Max Marks	Marks Awarded
O1	Why must you wear an anti-static wrist strap before touching internal parts?	To discharge static electricity that can damage sensitive components such as RAM, CPU and motherboard; strap earths the body (3: reason, effect, how strap helps)	3	
O2	State the function of the motherboard, the CPU and the RAM.	Motherboard connects all components; CPU processes instructions; RAM temporarily holds data and programs in use (1 mark each)	3	
O3	Name three safety precautions you take before opening a system unit.	Shut down; unplug power and cables; use anti-static strap; work on a clean dry surface; use correct tools (any 3)	3	
O4	Why is it important to clean the fan and the inside of the case?	Dust blocks airflow, causes overheating, noise, slowdowns and shutdowns, and shortens component life (3: two reasons and a consequence)	3	
O5	Describe what happens during POST.	BIOS tests CPU, RAM, video, keyboard and storage; reports errors by beeps/messages; then hands over to boot loader to load OS (3: tests, errors, handover)	3	
O6	Differentiate between a cold boot and a warm boot, and give when each is used.	Cold: starts from power off, full POST, used after shutdown or hardware changes. Warm: restart without powering off, may skip full POST, used after software changes or freezing (3)	3	
O7	Name three rear ports, and for each state the device that connects to it.	e.g. USB - mouse/printer; HDMI/VGA - monitor/projector; audio jack - speakers; RJ-45 - network (1 mark per correct pair)	3	
O8	What would you do if the computer gives beep codes and does not display anything on start-up?	Count/identify beep pattern, check BIOS manual; reseat RAM, video card and cables; check power; test with known good parts (3: identify, reseat/check, test)	3	
O9	Explain the steps for installing and configuring a printer.	Connect cable and power; install driver; set as default; print a test page; troubleshoot paper/toner if needed (3)	3	

No.	Oral Question	Expected Answer (Marking Guide)	Max Marks	Marks Awarded
O10	How do you display your screen through a projector, and what settings can you change?	Connect VGA/HDMI; select correct input on projector; press Windows+P or Fn key; choose Duplicate/Extend/Second screen only; adjust resolution and refresh rate (3)	3	
	Total Oral Marks		**30**	

Section F: Marks Summary and Decision Outcome

Component	Max Marks	Marks Awarded	Percentage
Practical Checklist	50		
Product Checklist	5		
Oral Checklist	30		
Grand Total	**85**		

Overall percentage = (Total awarded / 85) x 100

Decision Outcome

The pass mark is set at 60% (51 out of 85). The assessing institution may change this threshold to match its policy.

Decision	Tick (✓)
COMPETENT (Total of 50 marks or more)	
NOT YET COMPETENT (Total below 49 marks)	

Assessor's Comments and Feedback

Strengths observed:

_____

_____

Areas needing improvement:

_____

_____

Recommendation (e.g. proceed, reassess in tasks ____):

_____

Section G: Declarations and Signatures

Candidate's declaration: I confirm that the assessment was conducted fairly, I understand the outcome, and I was informed of my right to appeal.

Role	Name	Signature	Date
Candidate			
Assessor			
Internal Verifier / Moderator			
Head of Department			

Appeal: The candidate may appeal in writing within the period allowed by the college's assessment policy.